

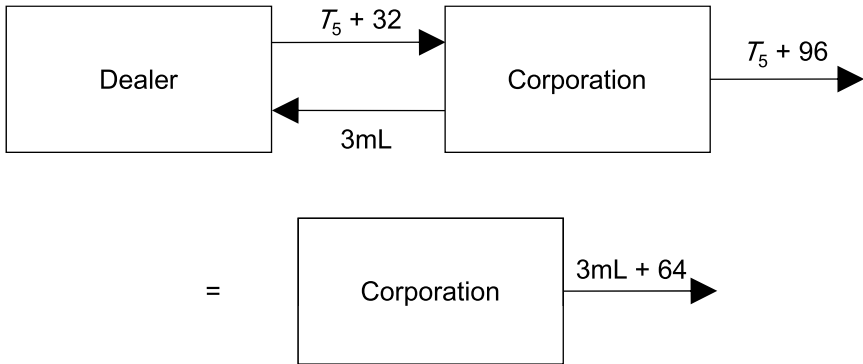


Figure S.2: Synthetic Floating Rate Funding Obtained by Corporation

3. a) Instead of paying fixed at the mid-market rate of 5.60%, the dealer has inadvertently paid at a rate of 5.65%. This mistake is against the dealer. We should use the PV Coupon 01 number in order to determine the loss to the dealer. The value of this mistake is

$$\$43,321 \times 5 \times 0.5 = \$108,302.$$

In the preceding equation, the factor of 5 is used since the mistake was 5 bps, and the PV Coupon 01 number supplied in Table 2.3 is for a 1 bp change. The factor of 0.5 in this equation accounts for the fact that the notional is \$50mm, not \$100mm as used in the calculations in Table 2.3.

b) Since the dealer is paying fixed and the market has rallied, the swap is in favor of the client and against the dealer. Since we are dealing with a change in the swap curve for a swap that has already been executed, it would be most appropriate to use the PV01 numbers in Table 2.3. The loss to the dealer is

$$\$40,237 \times 5 \times 0.5 = \$100,592.$$

Notice we have used the PV01 number assuming the first LIBOR has already set; a rally in the market would not change the initial LIBOR set in the trade. Notice that the answer to this question is slightly different to the answer obtained in part a) due to the (appropriately) different PV01 methodology employed.

Chapter 3

2. a) As shown in the top half of Figure S.3, the 2x5 swap rate is 7.071%. As a handcheck to this calculation, consider receiving in a 5-year swap and paying in a 2-year swap. The carry over the first two years is $6.60\% - 6.00\% = 60$ bps. The future value of this carry (using the 2-year swap rate as discount rate) is 123.6 bps. Amortizing this over the back three years (using the 5-year swap rate as discount rate) gives an annuity of 46.8 bps. Adding this annuity to the 5-year swap rate gives $6.60\% + 0.468\% = 7.068\%$, three tenths of a bp away from the true answer.

b) The value of the floating side of a swap including fictitious principal is par on any reset date. But today is not a reset date in the swap. The first reset date is in two years. Thus, we know that in two years the value of the floating side will be par. Therefore, the value of the floating side today is the present value of \$100mm to be received in two years:

$$\frac{\$100\text{mm}}{[1 + zc(2)]^2} = \frac{\$100\text{mm}}{(1.06012)^2} = \$88,979,417.$$

As shown in the bottom half of Figure S.3, this can also be verified by modeling the swap with the assumption that the counterparties exchange principal at maturity in five years. We find that the value of the fictitious floating rate bond is \$88,979,417.

3. a) As it stands today, the client is receiving 5.80% when the 5-year swap rate is 6.60%. As shown in the top half of Figure S.4, the mark-to-market on the trade is \$3,339,836 against the client (the trade is modeled from the perspective of the client). As a handcheck to this calculation, it is as if the client is paying an annuity of $6.60\% - 5.80\% = 80$ bps for five years. Put $\text{PMT} = 80$ bps, $n = 5$, and $i = 6.60\%$ and get $\text{PV} = 331.6$ bps of the notional, or \$3,315,590.

b) The client is currently receiving fixed. In order to unwind the trade, it is as if the client will be paying fixed to the dealer. Thus, it is the offer side 5-year swap rate that is applicable in the pricing of the unwind. The 5-year mid-market swap rate is 6.60%. The offer side 5-year swap rate is 1 bp from there, or 6.61%. The pricing of the unwind using a 5-year swap rate of 6.61% is shown in the bottom half of Figure S.4. The difference between the two NPVs, $\$3,381,267 - \$3,339,836 = \$41,431$, represents the profitability that the dealer makes on the unwind if it deals at the offer side rate.

SOLUTIONS TO SELECTED PROBLEMS

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	0	0	0	0
2	6.012%	6.426%	0	0	0	0
3	6.221%	6.640%	7,071,362	5,900,263	6,640,266	5,540,562
4	6.437%	7.086%	7,071,362	5,509,855	7,085,633	5,520,975
5	6.659%	7.553%	7,071,362	5,122,944	7,552,520	5,471,526
				16,533,062		16,533,062
Fixed rate			7.071%	NPV	0	
Notional			100,000,000			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	0	0	0	0
2	6.012%	6.426%	0	0	0	0
3	6.221%	6.640%	7,071,362	5,900,263	6,640,266	5,540,562
4	6.437%	7.086%	7,071,362	5,509,855	7,085,633	5,520,975
5	6.659%	7.553%	107,071,362	77,569,298	107,552,520	77,917,880
				88,979,417		88,979,417
Fixed rate			7.071%	NPV	0	
Notional			100,000,000			

Figure S.3: The 2x5 Swap Rate (*top half*) and with Fictitious Principal (*bottom half*)

6. a) The 2-year swap rate is given as 6%, and in problem 2 we found that the 2x5 swap rate is 7.071%.

b) The fixed rate in the 2-year swap will need to be increased, and the fixed rate in the 2x5 swap will need to be decreased. As we start increasing the fixed rate in the 2-year swap, the value of this trade moves in favor of the investor. As we start decreasing the fixed rate in the 2x5 swap, the value of this trade moves against the investor. In a calculation not shown here,

SOLUTIONS TO SELECTED PROBLEMS

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	5,800,000	5,492,424	(5,600,000)	(5,303,030)
2	6.012%	6.426%	5,800,000	5,160,806	(6,425,703)	(5,717,553)
3	6.221%	6.640%	5,800,000	4,839,454	(6,640,266)	(5,540,562)
4	6.437%	7.086%	5,800,000	4,519,237	(7,085,633)	(5,520,975)
5	6.659%	7.553%	5,800,000	4,201,889	(7,552,520)	(5,471,526)
				24,213,810		(27,553,645)
		Fixed rate	5.800%	NPV	(3,339,836)	
		Notional	100,000,000			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	5,800,000	5,492,424	(5,600,000)	(5,303,030)
2	6.012%	6.426%	5,800,000	5,160,806	(6,425,703)	(5,717,553)
3	6.221%	6.640%	5,800,000	4,839,454	(6,640,266)	(5,540,562)
4	6.437%	7.086%	5,800,000	4,519,237	(7,085,633)	(5,520,975)
5	6.670%	7.611%	5,800,000	4,199,617	(7,610,687)	(5,510,685)
				24,211,538		(27,592,805)
		Fixed rate	5.800%	NPV	(3,381,267)	
		Notional	100,000,000			

Figure S.4: Mark-to-Market on Existing Swap with 5.80% Coupon (*top half*) and Unwind Value of Swap with Bid/Offer Taken into Account (*bottom half*)

when we set the rates on the two trades equal to 6.60%, the NPV of one trade exactly offsets the other trade. In particular, with a rate of 6.60%, the 2x5 swap has a negative mark-to-market of \$1,102,058 for the investor, and the 2-year swap has a positive mark-to-market of \$1,102,058 for the investor.

c) The rate that we solved for in part b) was the 5-year swap rate. Receiving fixed on \$100mm 2-year swaps and simultaneously receiving fixed on \$100mm

2x5 swaps is essentially the same thing as receiving fixed on \$100mm 5-year swaps. Changing the fixed coupons as we did in part b) shows that this is indeed the case.

Chapter 4

1. a) The value of the cap is 266.1 bps of the notional, or around \$2,661,000, as shown in Figure S.5.

b) The average vol is found by constraining each of the base vols to be the same and solving for the vol that makes the value of the cap 266.1 bps, the same value as determined in part a). As shown in Figure S.6, the average vol is 14.66%.

c) The value of the floor is 224.3 bps, or around \$2,243,000, as shown in Figure S.7.

d) From cap-floor parity, a 5-year swap in which a counterparty pays a fixed rate of 6.50% is equal to a long position in a 6.50% cap and a short position in a 6.50% floor. Thus, the value of the swap is $\$2,660,619 - \$2,243,140 = \$417,479$ in favor of the fixed rate payer.

We can verify this result from the bootstrap method as shown in Figure S.8. The value of the swap to the fixed rate payer is $\$27,553,645 - \$27,136,166 = \$417,479$, as expected.

2. The dealer is paying $\min(L, 8\%)$. Notice that $\min(L, 8\%) = L - \max(0, L - 8\%)$. Thus, the dealer is paying LIBOR less the payout on an 8% cap. The fixed rate in the swap will be the par swap rate reduced by the amortized value of the cap.

We first need to value an 8% cap. The cap is worth 100.1 bps, as shown in the top half of Figure S.9. In order to solve for the fixed rate in the swap, we put in an upfront amount to account for the value of the cap that benefits the dealer. As shown in the bottom half of Figure S.9, the fixed rate in the swap is 6.36%.

Cap model									
							Forward		Caplet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Caplet	in dollars
1	5.600%	0x1	5.600%	n/a	n/a	n/a	0.000%	0.000%	0
2	6.012%	1x2	6.426%	13%	-0.023	-0.153	0.299%	0.266%	266,139
3	6.221%	2x3	6.640%	14%	0.207	0.009	0.591%	0.493%	493,322
4	6.437%	3x4	7.086%	16%	0.450	0.173	1.077%	0.839%	839,288
5	6.659%	4x5	7.553%	15%	0.650	0.350	1.466%	1.062%	1,061,870
Strike 6.500%			Notional		100,000,000				2,660,619
									in bps 266.1

Figure S.5: Valuing the 5-Year 6.50% Cap

Cap model										
							Forward		Caplet	
Time	zc	Period	Forward	σ	d_1	d_2	premium	Caplet	in dollars	
1	5.600%	0x1	5.600%	n/a	n/a	n/a	0.000%	0.000%	0	
2	6.012%	1x2	6.426%	14.66%	-0.005	-0.152	0.342%	0.304%	303,992	
3	6.221%	2x3	6.640%	14.66%	0.207	-0.001	0.615%	0.514%	513,507	
4	6.437%	3x4	7.086%	14.66%	0.467	0.213	1.018%	0.793%	793,244	
5	6.659%	4x5	7.553%	14.66%	0.658	0.365	1.449%	1.050%	1,049,876	
Strike 6.500%			Notional		100,000,000				2,660,619	
									in bps	266.1

Figure S.6: Computing the Average Vol for the 5-Year 6.50% Cap

Floor model									
							Forward		Floorlet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Floorlet	in dollars
1	5.600%	0x1	5.600%	n/a	n/a	n/a	0.900%	0.852%	852,273
2	6.012%	1x2	6.426%	14.66%	-0.005	-0.152	0.416%	0.370%	370,101
3	6.221%	2x3	6.640%	14.66%	0.207	-0.001	0.475%	0.396%	396,471
4	6.437%	3x4	7.086%	14.66%	0.467	0.213	0.432%	0.337%	336,932
5	6.659%	4x5	7.553%	14.66%	0.658	0.365	0.397%	0.287%	287,363
Strike 6.500%			Notional		100,000,000				2,243,140
									in bps 224.3

Figure S.7: Valuing the 5-Year 6.50% Floor

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	(6,500,000)	(6,155,303)	5,600,000	5,303,030
2	6.012%	6.426%	(6,500,000)	(5,783,662)	6,425,703	5,717,553
3	6.221%	6.640%	(6,500,000)	(5,423,526)	6,640,266	5,540,562
4	6.437%	7.086%	(6,500,000)	(5,064,662)	7,085,633	5,520,975
5	6.659%	7.553%	(6,500,000)	(4,709,013)	7,552,520	5,471,526
				(27,136,166)		27,553,645
		Fixed rate	6.500%	NPV	417,479	
		Notional	100,000,000			

Figure S.8: Verifying Cap-Floor Parity Holds

Cap model									
							Forward		Caplet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Caplet	in dollars
1	5.600%	0x1	5.600%	n/a	n/a	n/a	0.000%	0.000%	0
2	6.012%	1x2	6.426%	13%	-1.621	-1.751	0.018%	0.016%	15,680
3	6.221%	2x3	6.640%	14%	-0.842	-1.040	0.134%	0.112%	111,784
4	6.437%	3x4	7.086%	16%	-0.299	-0.577	0.452%	0.352%	352,133
5	6.659%	4x5	7.553%	15%	-0.042	-0.342	0.720%	0.522%	521,865
Strike 8.000%			Notional		100,000,000				1,001,462
									in bps 100.1

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.600%	5.600%	6,360,117	6,022,838	(5,600,000)	(5,303,030)
2	6.012%	6.426%	6,360,117	5,659,195	(6,425,703)	(5,717,553)
3	6.221%	6.640%	6,360,117	5,306,809	(6,640,266)	(5,540,562)
4	6.437%	7.086%	6,360,117	4,955,668	(7,085,633)	(5,520,975)
5	6.659%	7.553%	6,360,117	4,607,673	(7,552,520)	(5,471,526)
				26,552,184		(27,553,645)
					Cap	1,001,462
Fixed rate			6.360%			
Notional			100,000,000	NPV	0	

Figure S.9: Calculating the Fixed Rate in the Swap with an Embedded Cap

Chapter 5

3. a) The answer is, it depends. First, suppose that the swap curve is as follows:

Maturity	Swap Rate
1 year	5.00%
2 year	5.11%
3 year	5.30%
4 year	5.47%
5 year	5.60%

In calculations not shown here, the 4-year swap rate, one year forward is equal to 5.772%, and the 3-year swap rate, two years forward is equal to 5.976%. Assume that the lognormal implied vol of both swaptions is 20%. As shown in Figure S.10, using Black's formula the value of a 1 into 4 7% receiver is 446.5 bps, and the value of a 2 into 3 7% receiver is 327.7 bps. Thus, the 1 into 4 is worth more than the 2 into 3.

Second, suppose that the swap curve is as follows:

Maturity	Swap Rate
1 year	1.00%
2 year	7.00%
3 year	7.00%
4 year	7.00%
5 year	7.00%

In calculations not shown here, the 4-year swap rate, one year forward is equal to 8.884%, and the 3-year swap rate, two years forward is equal to 7%. Assume that the lognormal implied vol of both swaptions is 20%. As shown in Figure S.11, using Black's formula the value of a 1 into 4 7% receiver is 28.3 bps, and the value of a 2 into 3 7% receiver is 179.7 bps. Thus, the 2 into 3 is worth more than the 1 into 4. This example was constructed such that the 1x5 swap rate is very high, making the value of the 1 into 4 receiver swaption relatively cheap.

5. In a calculation not shown here, the 2x5 forward swap rate is 7.595%. The appropriate vol to use is 26.5%.

Swaption model		1	into	4		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.8648
2	5.110%	5.113%	1x2	0.9051	d_2	-1.0648
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074		
5	5.600%	5.635%	4x5	0.7602		
F	5.772%		$A =$	3.3289		
σ	20.0%				$\chi N(-d_2) - FN(-d_1)$	0.013412
T	1		Notional	100,000,000	Receiver swaption	4,464,748
χ	7.000%				(in bps)	446.5

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.4180
2	5.110%	5.113%	1x2		d_2	-0.7009
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074		
5	5.600%	5.635%	4x5	0.7602		
F	5.976%		$A =$	2.4238		
σ	20.0%				$\chi N(-d_2) - FN(-d_1)$	0.013522
T	2		Notional	100,000,000	Receiver swaption	3,277,382
χ	7.000%				(in bps)	327.7

Figure S.10: An Example in which a 1 into 4 7% Receiver is Worth More Than a 2 into 3 7% Receiver